

Mobile Data Usage & Bill Management System

1. Introduction

The Mobile Data Usage & Bill Management System is a console-based Java application developed using Core Java, JDBC, and MySQL. The system tracks mobile data usage, calculates extra charges, and manages bill payments efficiently.

2. Objectives

- Record daily mobile data usage
- Calculate extra charges for excess usage
- Store and retrieve data using MySQL
- Understand JDBC and DAO–DTO architecture

3. Technologies Used

Programming Language: Java
Database: MySQL
Connectivity: JDBC
IDE: Eclipse / IntelliJ / NetBeans

4. System Architecture

DTO – DAO – DAO Implementation – Service – Main Class layered architecture.

5. Database Design

Database: mobile_db
Table: data_usage (id, mobile_no, data_used, extra_charge, payment_status)

6. Modules

DataUsageDTO – holds data
DataUsageDAO – interface for DB operations
DataUsageDAOImpl – JDBC logic
DataUsageService – business logic
MainApp – user interaction

7. Business Logic

Daily limit: 1.5GB
Extra charge: ₹50 per GB above limit
Initial status: UNPAID

8. Advantages

- Simple and user-friendly
- Real-time billing logic

- Easy to enhance

9. Limitations

- Console-based
- No security
- Fixed tariff

10. Future Enhancements

- GUI interface
- Monthly reports
- Authentication

11. Conclusion

This project demonstrates Java–MySQL integration using JDBC and DAO architecture, suitable for academic mini projects.